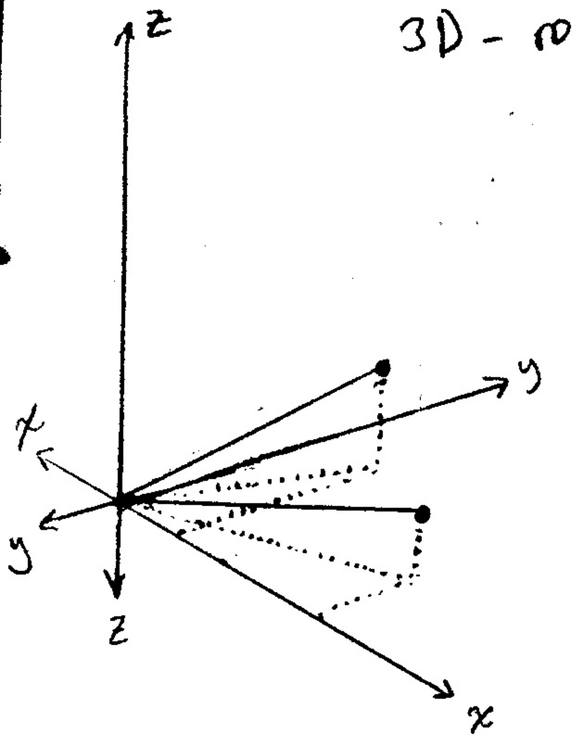
3D - rotation



* In theory for the free 3D rotation you need rotation a long one plane and then rotation a long another

* I have done this before but not on pipe, copy and pasting in desmos

* There are 6 possible combinations of planes but i'll just use xy and xz

$$x' = (x\cos(\phi) - y\sin(\phi))\cos(\theta) - z\sin(\theta)$$
$$y' = x\sin(\phi) + y\cos(\phi)$$
$$z' = x\sin(\theta) + z\cos(\theta)$$

* ϕ and θ are 2 different defined angles

Alternative x'

$$(x\cos(\phi) - z\sin(\phi))\cos(\theta) - y\sin(\theta)$$

$$z' = x\sin(\theta) + z\cos(\theta)$$
$$y' = x\sin(\theta) + y\cos(\theta)$$

$$x' = \begin{cases} x\cos(\theta) - z\sin(\theta) \\ x\cos(\theta) - y\sin(\theta) \end{cases}$$

* In most 3D surface equations replacing x, y, and z by x', y', and z' (the equation equivalents) allows you to rotate the surface by angles ϕ and θ

* In this case θ is pitch (Y-axis) rotation and ϕ is Yaw (Z-axis) rotation